

Practice Question LA -2

Saturday, 31 January 2026 10:14 am

Linear Combination & Span

Express the following as linear combinations of $\mathbf{u} = (2, 1, 4)$, $\mathbf{v} = (1, -1, 3)$, and $\mathbf{w} = (3, 2, 5)$.

- a.** $(-9, -7, -15)$ **b.** $(6, 11, 6)$ **c.** $(0, 0, 0)$

In each part, determine whether the vectors span R^3 .

- a.** $\mathbf{v}_1 = (2, 2, 2)$, $\mathbf{v}_2 = (0, 0, 3)$, $\mathbf{v}_3 = (0, 1, 1)$
b. $\mathbf{v}_1 = (2, -1, 3)$, $\mathbf{v}_2 = (4, 1, 2)$, $\mathbf{v}_3 = (8, -1, 8)$

Suppose that $\mathbf{v}_1 = (2, 1, 0, 3)$, $\mathbf{v}_2 = (3, -1, 5, 2)$, and $\mathbf{v}_3 = (-1, 0, 2, 1)$. Which of the following vectors are in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?

- a.** $(2, 3, -7, 3)$ **b.** $(0, 0, 0, 0)$
c. $(1, 1, 1, 1)$ **d.** $(-4, 6, -13, 4)$

In each part, determine whether the matrices span M_{22} .

- a.** $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

Determine whether the following polynomials span P_2 .

$$\begin{aligned} \mathbf{p}_1 &= 1 - x + 2x^2, & \mathbf{p}_2 &= 3 + x, \\ \mathbf{p}_3 &= 5 - x + 4x^2, & \mathbf{p}_4 &= -2 - 2x + 2x^2 \end{aligned}$$

Linearly Independent & Dependent

Explain why the following form linearly dependent sets of vectors. (Solve this problem by inspection.)

a. $\mathbf{u}_1 = (-1, 2, 4)$ and $\mathbf{u}_2 = (5, -10, -20)$ in R^3

b. $\mathbf{u}_1 = (3, -1)$, $\mathbf{u}_2 = (4, 5)$, $\mathbf{u}_3 = (-4, 7)$ in R^2

c. $\mathbf{p}_1 = 3 - 2x + x^2$ and $\mathbf{p}_2 = 6 - 4x + 2x^2$ in P_2

d. $A = \begin{bmatrix} -3 & 4 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 \\ -2 & 0 \end{bmatrix}$ in M_{22}

In each part, determine whether the vectors are linearly independent or are linearly dependent in R^3 .

a. $(-3, 0, 4)$, $(5, -1, 2)$, $(1, 1, 3)$

b. $(-2, 0, 1)$, $(3, 2, 5)$, $(6, -1, 1)$, $(7, 0, -2)$

In each part, determine whether the vectors are linearly independent or are linearly dependent in R^4 .

a. $(3, 8, 7, -3)$, $(1, 5, 3, -1)$, $(2, -1, 2, 6)$, $(4, 2, 6, 4)$

In each part, determine whether the three vectors lie in a plane in R^3 .

a. $\mathbf{v}_1 = (2, -2, 0)$, $\mathbf{v}_2 = (6, 1, 4)$, $\mathbf{v}_3 = (2, 0, -4)$

a. Show that the three vectors $\mathbf{v}_1 = (0, 3, 1, -1)$, $\mathbf{v}_2 = (6, 0, 5, 1)$, and $\mathbf{v}_3 = (4, -7, 1, 3)$ form a linearly dependent set in R^4 .

b. Express each vector in part (a) as a linear combination of the other two.

For which real values of λ do the following vectors form a linearly dependent set in R^3 ?

$$\mathbf{v}_1 = \left(\lambda, -\frac{1}{2}, -\frac{1}{2}\right), \quad \mathbf{v}_2 = \left(-\frac{1}{2}, \lambda, -\frac{1}{2}\right), \quad \mathbf{v}_3 = \left(-\frac{1}{2}, -\frac{1}{2}, \lambda\right)$$

Bases & Dimension

Show that the following polynomials form a basis for P_2 .

$$x^2 + 1, \quad x^2 - 1, \quad 2x - 1$$

In each part, show that the set of vectors is not a basis for $\mathbb{R}^3$.

a. $\{(2, -3, 1), (4, 1, 1), (0, -7, 1)\}$

b. $\{(1, 6, 4), (2, 4, -1), (-1, 2, 5)\}$

In Exercises **1–6**, find a basis for the solution space of the homogeneous linear system, and find the dimension of that space.

1.	$x_1 + x_2 - x_3 = 0$	2.	$3x_1 + x_2 + x_3 + x_4 = 0$
	$-2x_1 - x_2 + 2x_3 = 0$		$5x_1 - x_2 + x_3 - x_4 = 0$
	$-x_1 + x_3 = 0$		

The vectors $\mathbf{v}_1 = (1, -2, 3)$ and $\mathbf{v}_2 = (0, 5, -3)$ are linearly independent. Enlarge $\{\mathbf{v}_1, \mathbf{v}_2\}$ to a basis for $\mathbb{R}^3$.

The vectors $\mathbf{v}_1 = (1, 0, 0, 0)$ and $\mathbf{v}_2 = (1, 1, 0, 0)$ are linearly independent. Enlarge $\{\mathbf{v}_1, \mathbf{v}_2\}$ to a basis for $\mathbb{R}^4$.